Active Maintenance Report: coppock_ekins_kirby_2018

Coppock, Ekins and Kirby (2018, Quarterly Journal of Political Science): The Long-lasting Effects of Newspaper Op-Eds on Public Opinion

2026-08-01

Table of contents

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Coppock, Ekins and Kirby (2018), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1561/100.00016112
Replication archive	10.7910/DVN/3KBCJF
Pre-analysis plan	osf.io/rmkw3

The data are not redistributed here. The deposit is 1.4 MB across 20 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_ekins_kirby_2018.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 20 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, estimatr, lmtest, modelsummary, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about thirty seconds. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Almost. Fifteen of the sixteen deposited R scripts execute without error on a current R installation. The sixteenth, `opeds_het_fx_analysis.R`, calls `waldtest()` without ever loading `lmtest`; the package installs from CRAN and one added `library(lmtest)` line fixes it. Every other package the archive names is still on CRAN and still works, including `coefplot`, whose `position_dodgev()` the figure code depends on.

Running is not the same as reproducing, and here the difference is large. Of the 502 published quantities that can be checked against a script, 20 no longer reproduce. 17 of those 20 are in Table 9, which has only eighteen cells. The archive computes those joint F tests as `waldtest(lm_robust(...), lm_robust(...), test = "F")`. When the archive was written, `lmtest::waldtest()` had no robust variance to take from an `lm_robust` object and silently fell back to the classical F test, which is what the article prints. Current `estimatr` supplies the variance, so the identical line now returns an HC2 Wald test instead. Every cell of the table moves. The one cell that still matches to three decimals (MTurk / Flat Tax / Main DV, 0.009) matches by rounding.

No conclusion in the paper turns on the difference. The two tests the article calls significant, Amtrak and Flat Tax on MTurk, stay significant under HC2; the fourteen it calls null stay null.

The remaining 3 failures are not the archive's fault at all. The article's prose reports 3,567 MTurk subjects, 2,169 elite subjects and 1,349 elite Wave 2 responses; its own Tables 1 and 5, the deposited data, and both the archive and the rewrite give 3,571, 2,181 and 1,358. Those are typesetting or transcription slips in the text, not analysis errors, and the tables next to them are right.

1.2 Does the maintained rewrite reproduce the paper?

Yes, on everything the article states. All 499 of the 502 verifiable ground truth claims match the published values to reported precision, including the Table 9 p-values, which the rewrite reproduces by reporting the classical F test alongside the HC2 one rather than by choosing between them silently. The 3 claims recorded as failures are the text-versus-table discrepancies above, where the rewrite agrees with the article's tables and disagrees with its prose. 4 further quantities are computed and committed but never printed in the article, so they are marked unverifiable rather than matched.

2 Paper overview

Citation: Coppock, A., Ekins, E. and Kirby, D. (2018). "The long-lasting effects of newspaper op-eds on public opinion." *Quarterly Journal of Political Science*, 13(1), 59-87. DOI:

10.1561/100.00016112

Summary: Two randomized panel survey experiments estimate the persuasive effect of real newspaper opinion pieces on policy attitudes. One runs on Amazon Mechanical Turk (N = 3,571), the other on a convenience sample of political and policy elites (N = 2,181) recruited from a list of 32,498 email addresses. Subjects were assigned to read one of five op-eds, on Amtrak, climate change, the flat tax, veterans' health care and Wall Street regulation, or to a control condition that read nothing; the elite study dropped the climate arm for want of sample. Outcomes were measured immediately, ten days later, and, on MTurk only, thirty days later. The op-eds moved attitudes on their target issues by roughly 0.4 to 0.9 points on seven-point scales and 0.3 to 0.7 standard deviations on composite scales, moved attitudes on non-target issues essentially not at all, produced effects that were still visible thirty days later, and produced smaller effects among elites than among the mass public.

3 Original archive reproducibility

The deposit is twenty files: three `.RData` data files, sixteen R scripts and a README.

Table 1: Deposited scripts and their status under R 4.6.0.

Script	Produces	Status
opeds_source.R	Helper functions, sourced by every other script	Clean
opeds_recontact_rates.R	Tables 1 and 5, plus the recontact rates on pp. 66 and 72	Clean
opeds_main_analysis.R	Tables 3, 4, 6 and 7	Clean
opeds_compare_samples.R	Table 8	Clean
opeds_het_fx_analysis.R	Figure 1, Table 9, and the by-party averages on p. 77	Errors: waldtest() called without
opeds_persistence.R	Figure 2	Clean
opeds_agreement.R	Table 11	Clean
opeds_compliance.R	Reading-time claims on pp. 69 and 72	Clean
opeds_monkey.R	A presentation figure that is in neither the article nor the appendix	Clean
opeds_het_fx_age_analysis.R	Figure I.10 (appendix)	Clean
opeds_persistence_pid.R	Figure H.9 (appendix)	Clean
opeds_appendix_attrition.R	Tables D.4 to D.7 (appendix)	Clean
opeds_appendix_demographics.R	Tables B.1 to B.3 and Figure B.6 (appendix)	Clean
opeds_appendix_distractor.R	Figure C.8 (appendix)	Clean
opeds_appendix_heatmaps.R	Figures A.1 to A.5 (appendix)	Clean
opeds_appendix_mj.R	Figure C.7 (appendix)	Clean

Four things are worth recording beyond the single error.

The one error. `opeds_het_fx_analysis.R` loads `tidyverse`, `estimatr`, `coefplot` and `reshape2`, then calls `waldtest()` at lines 171 and 200. `waldtest()` belongs to `lmtest`, which is never loaded. The script builds Figure 1 and prints the by-party averages before it dies, so the failure is at the F tests only. Adding `library(lmtest)` is the whole fix.

The archive's own README undercounts itself. It states that the archive "has 15 scripts" and lists fifteen. The deposit contains sixteen R files. `opeds_monkey.R` appears in neither the list nor the count; it draws a bar chart of the percentage agreeing with each op-ed author that appears in no part of the published article or appendix.

The archive writes into itself. Every script begins with `rm(list = ls())` and a comment instructing the user to set the working directory to the archive folder. Running them there leaves an `Rplots.pdf` behind, which is how the copy of this deposit that seeded this repository came to contain a file the deposit does not. All twenty deposited files verify byte for byte against Dataverse; the stray plot file has been removed, so `original/` now holds the deposit and nothing else.

Table 7's significance stars are wrong in the archive. The three other `stargazer()` calls in `opeds_main_analysis.R` pass `p = starprep(..., stat = "p.value")`. The fourth, the one that makes Table 7, omits `stat = "p.value"`, so `stargazer` receives standard errors where it expects p-values and marks every coefficient in the table with a single star. The published Table 7 carries the correct stars, so the version that reached print was not the version the deposit reproduces. Estimates and standard errors are unaffected, and the rewrite's `modelsummary` tables take stars from p-values.

4 The Table 9 drift

This is the substantive reproducibility finding, and it is worth stating in full because the mechanism is invisible from the code.

`opeds_het_fx_analysis.R` tests whether treatment effects differ across Democrats, Independents and Republicans by comparing `dv ~ Z + pid_3_cat` against `dv ~ Z * pid_3_cat`:

```
fit_r <- lm_robust(local_formula_r, data = local_df)
fit_u <- lm_robust(local_formula_u, data = local_df)
anova_fit <- waldtest(fit_r, fit_u, test = "F")
```

`lmtest::waldtest()` looks for a `vcov` method on the objects it is given. Older `estimatr` did not export one for `lm_robust`, so `waldtest()` fell through to the classical variance implied by the residuals, and the p-values the article prints are classical F tests. Current `estimatr` does export it, so the same three lines now produce an HC2 Wald test. The author's evident intent, visible in the choice of `lm_robust`, was the robust test; the published numbers are the classical one.

Table 2: Table 9 as published, as the rewrite reproduces it, and as the archive's own code now returns it.

Sample	Issue	DV	Published	Classical F	HC2 Wald

MTurk	Amtrak	Main DV	0.045	0.045	0.042
MTurk	Amtrak	Scale DV	0.029	0.029	0.020
MTurk	Climate	Main DV	0.845	0.845	0.883
MTurk	Climate	Scale DV	0.243	0.243	0.362
MTurk	Flat Tax	Main DV	0.009	0.009	0.009
MTurk	Flat Tax	Scale DV	0.001	0.001	0.000
MTurk	Veterans	Main DV	0.878	0.878	0.876
MTurk	Veterans	Scale DV	0.562	0.562	0.615
MTurk	Wall Street	Main DV	0.310	0.310	0.341
MTurk	Wall Street	Scale DV	0.371	0.371	0.374
Elite	Amtrak	Main DV	0.436	0.436	0.487
Elite	Amtrak	Scale DV	0.342	0.342	0.383
Elite	Flat Tax	Main DV	0.146	0.146	0.187
Elite	Flat Tax	Scale DV	0.293	0.293	0.284
Elite	Veterans	Main DV	0.434	0.434	0.431
Elite	Veterans	Scale DV	0.291	0.291	0.320
Elite	Wall Street	Main DV	0.316	0.316	0.372
Elite	Wall Street	Scale DV	0.265	0.265	0.283

The rewrite computes both and writes both to `maintained/output/table_9_het_fx_ftests.csv`. The `.tex` table it renders uses the classical column, because that is the published table. `party_ftest()` in `helpers.R` takes `se_type` as a required argument for the same reason: an analysis choice this consequential should not sit in a default.

5 Errata

None. The rewrite changes no analytical decision. The three ground truth rows recorded as `match_rewrite = 0` are places where the article's prose disagrees with the article's own tables, listed below, and no correction to the code would resolve them.

Table 3: Text-versus-table discrepancies in the published article.

Claim	In the text	In the data	Note
MTurk subjects enrolled (p. 66)	3567	3571	Article says 3,567; Table 1 and the data both give 3,571. Text typo, c
Elite subjects who completed the survey (p. 72)	2169	2181	Article says 2,169; Table 5 and the data both give 2,181. Same class c
Elite complete responses in Wave 2 (p. 72)	1349	1358	Article says 1,349; Table 5 and the data both give 1,358.

The MTurk discrepancy was already known; the two elite ones are new. All three run the same way: the prose undercounts, the table is right, and the analyses use the table's number.

6 Ground truth summary

`ground_truth/coppock_ekins_kirby_2018_ground_truth.csv` records 506 published quantities. `value_paper` was read from the article PDF and from nowhere else. `value_script` is what the deposited scripts return today; `value_rewrite` is what `maintained/output/` contains.

Table 4: Ground truth coverage by published table.

Table or claim	Claims	Archive matches	Archive fails	Rewrite matches	Rewrite fails	Unverifiable
table_1	49	49	0	49	0	0
table_3	70	70	0	70	0	0
table_4	70	70	0	70	0	0
table_5	30	30	0	30	0	0
table_6	48	48	0	48	0	0
table_7	48	48	0	48	0	0
table_8	88	88	0	88	0	0
table_9	18	1	17	18	0	0
table_11	66	66	0	66	0	0
text	19	12	3	12	3	4

Coverage is every estimate, standard error, constant, R-squared and N printed in Tables 1, 3, 4, 5, 6, 7, 8, 9 and 11, plus the in-text quantities on pp. 66, 69, 72 and 77. Table 2 lists the op-eds and Table 10 is a hand-assembled circulation table built from figures the archive does not contain, so neither is reproducible from the deposit and neither is claimed here. Figures 1 and 2 print no numbers; their underlying estimates are written to CSV so they can be diffed, but there is nothing in them to check against the page.

7 Maintained rewrite

Eleven scripts and a `helpers.R`, one per published float, plus three for the in-text quantities.

Table 5: Scripts in maintained/.

Script	Produces
helpers.R	Packages, deposited data, shared labels and three shared estimators
table_1_5_design.R	Tables 1 and 5
table_3_4_mturk_effects.R	Tables 3 and 4
table_6_7_elite_effects.R	Tables 6 and 7
table_8_compare_samples.R	Table 8
table_9_het_fx_ftests.R	Table 9, both F tests
table_11_agreement.R	Table 11 and its precision-weighted averages
figure_1_het_fx_by_party.R	Figure 1 and the by-party estimates behind it
figure_2_persistence.R	Figure 2 and the group means behind it
text_recontact_rates.R	Enrolment, recontact and chi-square claims, pp. 66 and 72
text_partisan_pw_averages.R	By-party averages and the Republican-Democrat difference, p. 77

Substitutions. `stargazer` plus `sink()` becomes `modelsummary(output = ...)`, which writes straight to a file and takes its stars from p-values. `reshape2::melt()` and `dcast()` become `pivot_longer()` and `pivot_wider()`. `coefplot::position_dodgev()` and `geom_errorbarh()` become `position_dodge()` and `geom_linerange()`, since ggplot2 4.x dodges along a discrete y axis without help. `do(tidy(...))` becomes `reframe(tidy(...))`. `lm()` plus `estimatr::starprep()` becomes `lm_robust(se_type = "HC2")`, which is the same variance estimator reached directly. `xtable` is left behind rather than replaced, because the tables it built are hand-assembled and `knitr::kable` plus `kableExtra` renders them straight into this report. No seeds are set anywhere in the archive and none are needed: nothing here samples.

Nothing typed in. The archive's `opeds_compliance.R` ends with two bare subtractions, `160.4120 - 134.00352` and `232.0353 - 186.43233`, whose inputs were copied from a previous run of the same script. The rewrite computes those differences from the data. They come to 26.41 and 45.60 seconds, which is where the article's "25-45 seconds less" comes from.

A finding that fell out of computing it. The article says the elite sample "spent 25-45 seconds less time reading our treatment articles than the MTurk sample." Computed across all four arms the two samples share, rather than the two the archive typed in, the differences are:

Table 6: Trimmed mean reading time by arm, MTurk minus elite.

Op-ed	MTurk (s)	Elite (s)	Difference (s)
Amtrak	160.4	134.0	26.4
Flat Tax	232.0	186.4	45.6
Veterans	176.6	207.3	-30.7
Wall Street	108.9	80.4	28.5

Three arms fall in or near the stated range. On the veterans op-ed the elite sample spent thirty-one seconds more, not less. The claim as written holds for three of four op-eds.

A defect found in this rewrite and fixed. `table_11_agreement.R` passed the treatment arm to its fitting function from inside a `mutate()`, where the bare name `arm` resolves to the whole column rather than the current row. Every model was therefore fit on all six arms at once. The script produced 41 rows where the published table has 27, labelled them with repeating topic names, and computed precision-weighted averages of 0.05 and 0.03 where the article reports 0.20 and 0.12. Nothing in the pipeline caught it, because the ground truth had no rows for Table 11 to catch it against. It has 66 now.

8 Figure verification

Neither published figure prints a number, so verification is visual against the article plus a numeric diff of the estimates behind each panel.

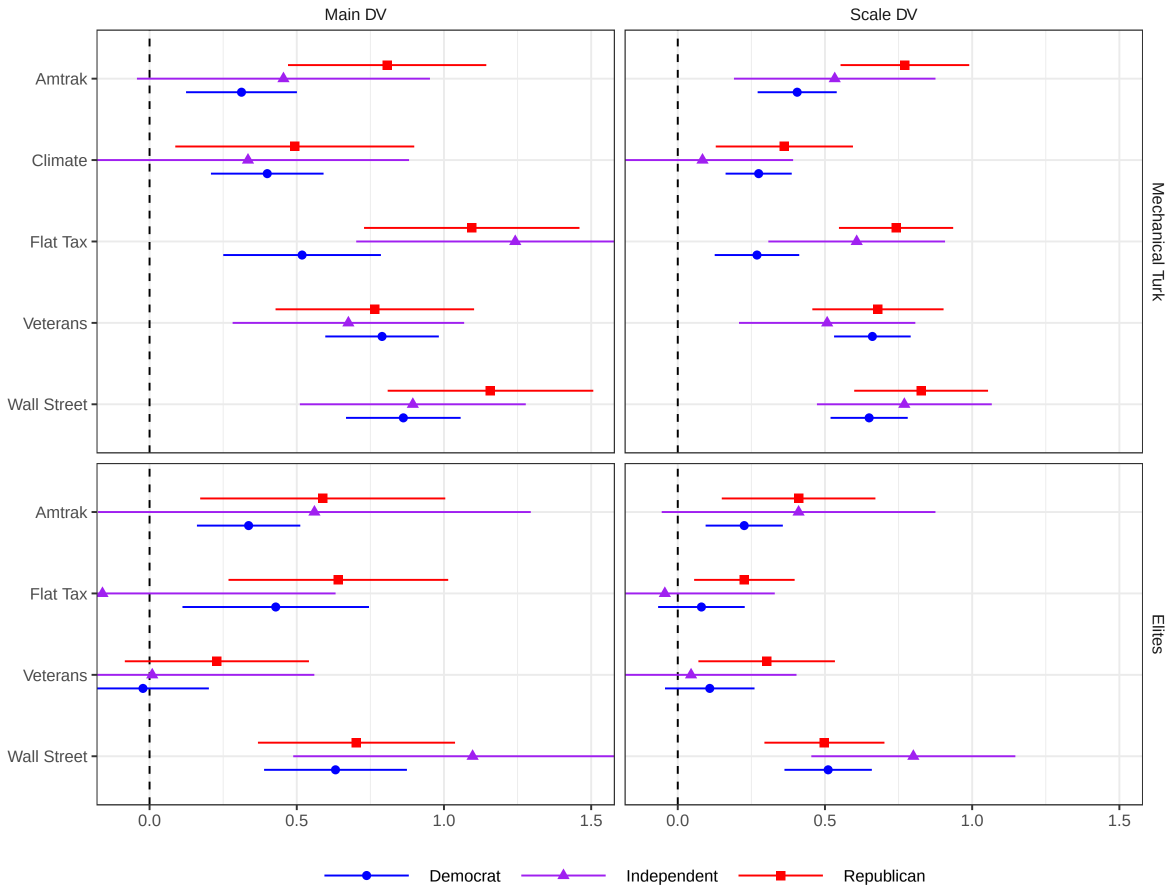

Figure 1, effects of treatment by party and experimental sample. The rewrite reorders the issue axis so the panels read top to bottom in the article's order and drops `coord_flip()`, which the current ggplot2 does not need.

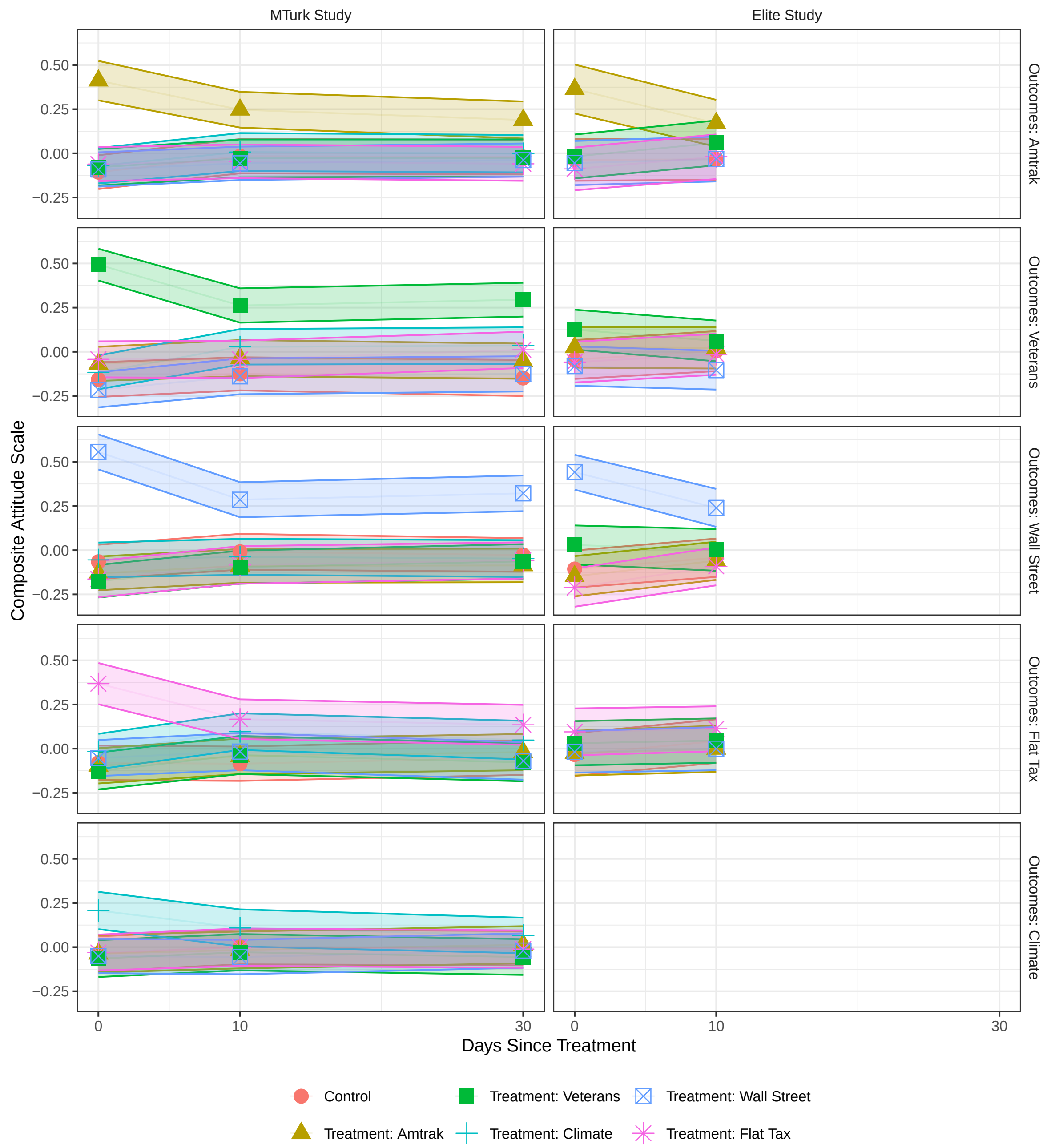

Figure 2, long-term effects of treatment. The archive plots the wave index against an axis relabelled 0, 10 and 30 days, which compresses the ten-day gap and the twenty-day gap to the same width. The rewrite plots the actual day counts.

9 Rewrite verification

Every script was run twice from a clean session and the outputs diffed. All twenty-eight CSV, TeX and PNG files are byte-identical between runs; the two PDF figures differ only in the creation timestamp a PDF records internally. Nothing in the rewrite samples, so no seed is involved in that result.

`run_all.R` was run end to end immediately before this report was rendered, so the committed `maintained/output/` is the product of a single ordered run rather than of scripts executed one at a time. The order matters in one place: `text_partisan_pw_averages.R` averages the estimates `figure_1_het_fx_by_party.R` writes, so it runs after it.

10 Checksums

All twenty deposited files were fetched from Dataverse with `?format=original` and hashed. The MD5 of what Dataverse serves agrees with the MD5 Dataverse publishes for all twenty, including `elite_opeds_cleaned.RData` and `mturk_opeds_cleaned.RData`, which Dataverse ingested to tabular `.tab` files: the published checksums describe the deposited RData, not the derived table. The local copy that seeded this repository is byte-identical to what the server returns, so nothing in it drifted in the eight years since deposit.

11 R environment

Table 7: Environment this report was rendered in.

Component	Version
R	4.6.0
tidyverse	2.0.0
estimatr	1.0.6
lmtest	0.9.40
modelsummary	2.6.0
knitr	1.51
kableExtra	1.4.0
here	1.0.2

The archive's own README names no R version. The packages it names are `coefplot`, `dplyr`, `estimatr`, `MASS`, `nnet`, `reshape2`, `stargazer`, `tidyverse` and `xtable`; `lmtest` is used but not named, which is the archive's one error.